Customer Churn Prediction: Develop a model to predict customer churn in a subscription-based business.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score,precision_score,recall_score,f1_score,confusion_matrix
import pickle
```

```
df=pd.read_csv('/content/drive/MyDrive/customer_churn_dataset.csv')
df.head()
```

	CustomerID	Gender	Age	Tenure_Months	SubscriptionType	MonthlyCharges	TotalCharges	PaymentMethod	SupportTickets	Co
0	CUST1000	Male	49	54	Standard	99.33	4912.77	Net Banking	1	
1	CUST1001	Female	56	8	Standard	12.01	604.59	Debit Card	2	
2	CUST1002	Male	66	27	Basic	57.62	2019.39	Cash	4	
3	CUST1003	Male	69	27	Premium	97.08	4848.88	Credit Card	7	
4	CUST1004	Male	49	34	Standard	54.69	4334.26	Credit Card	9	

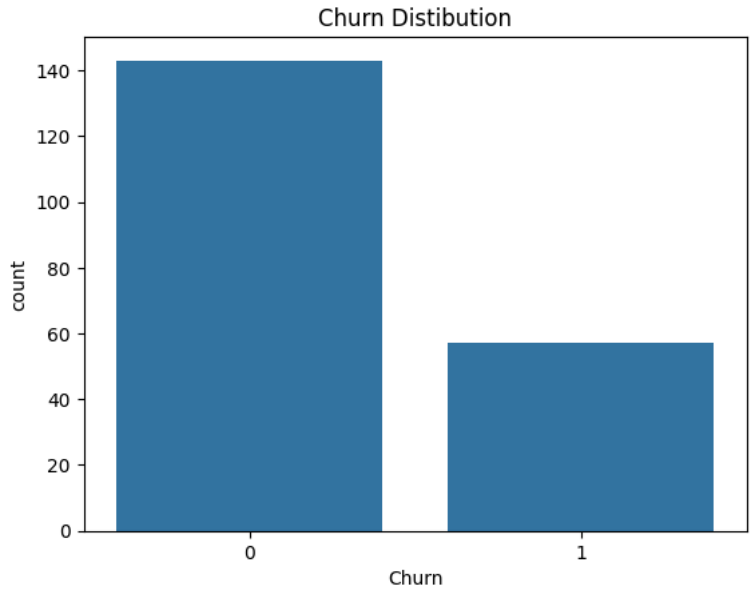
Next steps: Generate code with df | New interactive sheet

```
print("Shape:",df.shape)
df.info()
df.describe()
```

```
Shape: (200, 11)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 11 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   CustomerID        200 non-null    object
 1   Gender            200 non-null    object
 2   Age               200 non-null    int64
 3   Tenure_Months     200 non-null    int64
 4   SubscriptionType  200 non-null    object
 5   MonthlyCharges    200 non-null    float64
 6   TotalCharges      200 non-null    float64
 7   PaymentMethod     200 non-null    object
 8   SupportTickets    200 non-null    int64
 9   ContractType      200 non-null    object
 10  Churn             200 non-null    int64
dtypes: float64(2), int64(4), object(5)
memory usage: 17.3+ KB
```

	Age	Tenure_Months	MonthlyCharges	TotalCharges	SupportTickets	Churn
count	200.000000	200.000000	200.000000	200.000000	200.000000	200.000000
mean	44.405000	31.925000	51.348250	2536.105100	4.505000	0.285000
std	15.353816	16.390933	27.570371	1408.107184	2.967153	0.452547
min	18.000000	1.000000	6.380000	72.930000	0.000000	0.000000
25%	32.750000	21.000000	27.335000	1292.312500	2.000000	0.000000
50%	46.000000	32.000000	50.530000	2537.170000	4.000000	0.000000
75%	56.000000	46.250000	73.062500	3753.577500	7.000000	1.000000
max	69.000000	58.000000	99.970000	4984.530000	9.000000	1.000000

```
sns.countplot(x='Churn',data=df)
plt.title("Churn Distibution")
plt.show()
```




```
df=df.drop("CustomerID",axis=1)
df=pd.get_dummies(df,drop_first=True)
df.head()
```

	Age	Tenure_Months	MonthlyCharges	TotalCharges	SupportTickets	Churn	Gender_Male	SubscriptionType_Premium	Subscript
0	49	54	99.33	4912.77	1	0	True	False	
1	56	8	12.01	604.59	2	0	False	False	
2	66	27	57.62	2019.39	4	0	True	False	
3	69	27	97.08	4848.88	7	0	True	True	
4	49	34	54.69	4334.26	9	1	True	False	

Next steps: Generate code with df New interactive sheet

```
X=df.drop(["Churn"],axis=1)
y=df["Churn"]
X.head(),y.head()
```

```
(   Age  Tenure_Months  MonthlyCharges  TotalCharges  SupportTickets  \
 0   49            54           99.33       4912.77               1
 1   56             8           12.01        604.59               2
 2   66            27           57.62       2019.39               4
 3   69            27           97.08       4848.88               7
 4   49            34           54.69       4334.26               9

    Gender_Male  SubscriptionType_Premium  SubscriptionType_Standard  \
 0         True                     False                       True
 1        False                     False                       True
 2         True                     False                      False
 3         True                      True                      False
 4         True                     False                       True

    PaymentMethod_Credit Card  PaymentMethod_Debit Card  \
 0                      False                     False
 1                      False                      True
 2                      False                     False
 3                       True                     False
 4                       True                     False

    PaymentMethod_Net Banking  PaymentMethod_UPI  ContractType_One Year  \
 0                       True              False                  False
 1                      False              False                  False
 2                      False              False                  False
 3                      False              False                  False
 4                      False              False                   True

    ContractType_Two Year
 0                   True
 1                   True
 2                   True
 3                   True
 4                  False  ,
 0    0
```

```
1    0
2    0
3    0
4    1
Name: Churn, dtype: int64)
```

```
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=50)
print("Train size:",X_train.shape)
print("Test size:",X_test.shape)
```

```
Train size: (160, 14)
Test size: (40, 14)
```

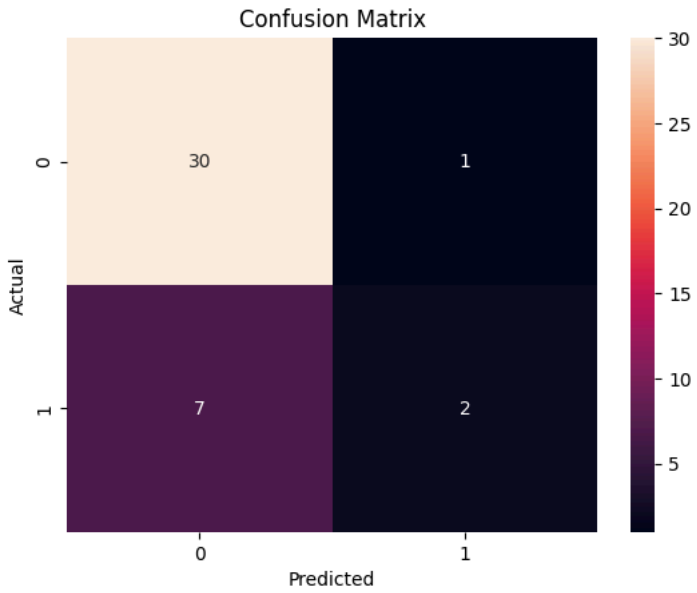
```
model=LogisticRegression(max_iter=1000)
model.fit(X_train,y_train)
with open("model","wb") as f:
    pickle.dump(model,f)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

```
y_pred=model.predict(X_test)
y_pred
```

```
array([0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0])
```

```
cm=confusion_matrix(y_test,y_pred)

sns.heatmap(cm,annot=True,fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```




```
print("Accuracy:",accuracy_score(y_test,y_pred))
print("Precision:",precision_score(y_test,y_pred))
print("Recall:",recall_score(y_test,y_pred))
print("F1 Score:",f1_score(y_test,y_pred))
```

```
Accuracy: 0.8
Precision: 0.6666666666666666
Recall: 0.2222222222222222
F1 Score: 0.3333333333333333
```